STUDYLAB HUB · COLLEGE CALCULUS I & AP CALCULUS · UNIT 3 (CHAPTER 3)

Unit 3: Applications of Derivatives & Optimization

Related Rates · Mean Value Theorem (MVT) · First & Second Derivative Tests · Concavity · Optimization & L'Hôpital's Rule

College Calculus I & AP Calculus — Unit 3 (Chapter 3): Related Rates, Linearization, L'Hôpital's Rule, MVT & Optimization

Independent, original study guide for College Calculus I & AP Calculus (informed by OpenStax open textbooks, CC BY 4.0).

1. Core Unit Roadmap

Section	Core Principle & Definitions	High-Yield Exam Application
Part 1 · Fundamental Concepts	Related Rates	Identifying core variables, boundary conditions, and conserved quantities
Part 2 · Quantitative Models & Mechanisms	Mean Value Theorem (MVT)	Applying governing equations and predicting how system state changes under perturbation
Part 3 · Advanced Problem Solving	Optimization & L'Hôpital's Rule	Avoiding common sign, unit, and conceptual traps on multiple-choice and free-response questions

2. Essential Principles & Step-by-Step Analysis

1. **First-Principles Mechanism**:

- Every problem in **Unit 3 (Chapter 3): Related Rates, Linearization, L'Hôpital's Rule, MVT & Optimization** reduces to identifying the underlying structure, forces, rates, or invariants before plugging in numbers.
- Always state the governing principle explicitly at the start of any free-response justification.

2. **High-Yield Comparison & Edge Cases**:

- Distinguish between equilibrium/steady-state behavior and transient/limiting behavior.
- Verify dimensional consistency and limiting cases (e.g., as distance or concentration approaches zero or infinity).

3. **Worked Exam Strategy**:

- Write out the known quantities with units, select the exact relation connecting them, solve symbolically first, and then substitute numerical values.